



Module
Presenter's Manual
for
**Enterprise Application
Development using EJB**

***Effective from: June 2024
Ver. 1.0***



Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	June 2024	New	-

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1. Introduction

At the end of this course, students will be able to:

- Learn the architecture for creating and deploying of component-based business applications
- Learn Jakarta Enterprise Beans Local and Remote Business Interfaces
- Learn Jakarta Messaging (JMS), Jakarta Connectors (JCA), and transactions

2. Information on Session Allocation

Module	Online (No. of Hours)
Enterprise Application Development using EJB	36

Throughout this Presenter's Manual, the module **Enterprise Application Development using EJB** will be referred to as **EJB**.

3. Module Deliverables available on OnlineVarsity

To aid the learning process, following are the deliverables

Student Deliverables:

1. Learner's Guide (eBook)

Resources available on OnlineVarsity for Students:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	FAQ - Student can access frequently asked questions and their answers.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

4. Week-wise Session Schedule

- A Session is of 2 hours duration.

➤ **Week-Wise Schedule**

Week	Day 1	Day 2	Day 3	Day 4
1	Session 1 EJB- TL1	Session 2 EJB- TL2	Session 3 EJB- TL3	Session 4 EJB- TL4
2	Session 5 EJB- TL5	Session 6 EJB- TL6	Session 7 EJB- TL7	Session 8 EJB- TL8
3	Session 9 EJB- TL9	Session 10 EJB- TL10	Session 11 EJB- TL11	Session 12 EJB- TL12
4	Session 13 EJB- TL13	Session 14 EJB- TL14	Session 15 EJB- TL15	Session 16 EJB- TL16
5	Session 17 EJB- TL17	Session 18 EJB- TL18		

TL: Online Session

5. Session Coverage

Sess No.	Session Title	Session Details	Deliverables' Mapping
1	EJB- TL1	All the topics as listed below from Session 1 and Session 2 of <i>Jakarta Enterprise Beans in Action</i> book should be covered in this session. <u>Session 1 – Introduction to Jakarta Enterprise Beans</u> ➤ Discuss an overview of Jakarta Enterprise Beans ➤ Describe an enterprise bean ➤ Explain the necessity for enterprise beans ➤ Describe how to set up the environment for Jakarta Enterprise Beans <u>Session 2 – Session Beans and their Types</u> ➤ Elaborate the types of session beans: stateful, stateless, and singleton ➤ Distinguish between stateful, stateless, and singleton sessions ➤ Describe entity beans ➤ Explain message-driven beans ➤ Explain message-driven controlled delivery: delivery active and delivery groups	<u>Jakarta Enterprise Beans in Action</u> SG - Session 1 & 2 XP - Session 1 & 2 TG- Session 1 & 2
2	EJB- TL2	All the topics as listed below from Session 3 of <i>Jakarta Enterprise Beans in Action</i> book should be covered in this session. <u>Session 3 – Resource Creation in Jakarta Enterprise Beans</u> ➤ Define Resource and Java Naming Directory Interface (JNDI) Naming ➤ Elaborate Data source, Objects, and Connection Pools ➤ Explain the creation of Resources administratively	<u>Jakarta Enterprise Beans in Action</u> SG - Session 3 XP - Session 3 TG- Session 3
3	EJB- TL3	All the topics as listed below from	<u>Jakarta Enterprise</u>

Sess No.	Session Title	Session Details	Deliverables' Mapping
		Session 4 of <i>Jakarta Enterprise Beans in Action</i> book should be covered in this session. <u>Session 4 – Working with Jakarta Enterprise Beans</u> ➤ Explain Transaction Processing with Enterprise beans ➤ Explain Concurrency Control in Enterprise applications ➤ Define Event-Driven Programming ➤ Describe Security in Enterprise applications	<u>Beans in Action</u> SG - Session 4 XP – Session 4 TG- Session 4
4	EJB– TL4	The Try It Yourself questions of Session 1 to Session 4 of <i>Jakarta Enterprise Beans in Action</i> book should be covered in this session.	<u>Jakarta Enterprise Beans in Action</u> Session 1-4
5	EJB– TL5	All the topics as listed below from Session 5 of <i>Jakarta Enterprise Beans in Action</i> book should be covered in this session. <u>Session 5 – Facelets in Jakarta Enterprise Beans</u> ➤ Define Facelets ➤ Explain lifecycle of Facelets ➤ Explain how to create a simple Facelet Application ➤ Define composite components ➤ Define Web Resources	<u>Jakarta Enterprise Beans in Action</u> SG - Session 5 XP – Session 5 TG- Session 5
6	EJB– TL6	All the topics as listed below from Session 6 of <i>Jakarta Enterprise Beans in Action</i> book should be covered in this session. <u>Session 6 – Jakarta Local and Remote Clients</u> ➤ Define Facelets ➤ Define Jakarta Local and Remote Client ➤ Explain when to use Local Client and Remote Client ➤ Explain the method of choosing	<u>Jakarta Enterprise Beans in Action</u> SG - Session 6 XP – Session 6 TG- Session 6

Sess No.	Session Title	Session Details	Deliverables' Mapping
		between Local and Remote Client ➤ Outline client view on Object lifecycle ➤ Describe object identity	
7	EJB- TL7	The Try It Yourself questions of Session 5 and Session 6 of <i>Jakarta Enterprise Beans in Action</i> book should be covered in this session.	<u>Jakarta Enterprise Beans in Action</u> Session 5 & 6
8	EJB- TL8	All the topics as listed below from Session 7 of <i>Jakarta Enterprise Beans in Action</i> book should be covered in this session. <u>Session 7 – Jakarta Messaging Services</u> ➤ Describe Jakarta Messaging ➤ Explain the goals of Jakarta Messaging, Jakarta Messaging Domains, and the importance of JMS ➤ Explain Jakarta Messaging Architecture, Jakarta Messaging ➤ Application, Jakarta Messaging APIs, classic API interfaces, and simplified API Interfaces ➤ Develop a Jakarta Messaging Application	<u>Jakarta Enterprise Beans in Action</u> SG - Session 7 XP – Session 7 TG- Session 7
9	EJB- TL9	All the topics as listed below from Session 8 of <i>Jakarta Enterprise Beans in Action</i> book should be covered in this session. <u>Session 8 – Understanding Jakarta Connectors Architecture</u> ➤ Define what is Jakarta Connectors ➤ Elaborate on JDBC connectors ➤ Outline the architecture of Jakarta Connectors ➤ Explain resource managers and connections ➤ Identify and describe	<u>Jakarta Enterprise Beans in Action</u> SG - Session 8 XP – Session 8 TG- Session 8

Sess No.	Session Title	Session Details	Deliverables' Mapping
		application components	
10	EJB- TL10	The Try It Yourself questions of Session 7 and Session 8 of <i>Jakarta Enterprise Beans in Action</i> book should be covered in this session.	<u>Jakarta Enterprise Beans in Action</u> Session 7 & 8
11	EJB- TL11	All the topics as listed below from Session 9 and Session 10 of <i>Jakarta Enterprise Beans in Action</i> book should be covered in this session. <u>Session 9 – Bean Validation</u> ➤ Explain Bean Validation ➤ Describe Repeating Annotations ➤ Explain Validating Constructors and Methods ➤ Explain Validating Null and Empty Strings ➤ Create Custom Constraints ➤ Describe Temporal Constraints Using Clock Provider ➤ Explain Built-In Value Extractors ➤ Customize Validator Messages <u>Session 10 – Execution of Enterprise Beans Using Transactions</u> ➤ Explain how to use transactions in Enterprise Beans ➤ Describe different events taking place ➤ Explain Java Naming and Directory Interface (JNDI) ➤ Elaborate on Remote/distributed execution	<u>Jakarta Enterprise Beans in Action</u> SG - Session 9 & 10 XP - Session 9 & 10 TG- Session 9 & 10
12	EJB- TL12	The Try It Yourself questions of Session 9 and Session 10 of <i>Jakarta Enterprise Beans in Action</i> book should be covered in this session.	<u>Jakarta Enterprise Beans in Action</u> Session 9 & 10
13	EJB- TL13	All the topics as listed below from	<u>Jakarta Enterprise</u>

Sess No.	Session Title	Session Details	Deliverables' Mapping
		Session 11 of <i>Jakarta Enterprise Beans in Action</i> book should be covered in this session. <u>Session 11 – Jakarta Context and Dependency Injection (CDI)-I</u> ➤ Describe an overview of Jakarta Context and Dependency Injection ➤ List and describe the benefits of Jakarta Context ➤ Explain how to enable Development Mode ➤ Identify the purpose of default Bean Discovery mode ➤ Outline annotations used in Bean Context ➤ Explain how to exclude Beans from the scanning process	<u>Beans in Action</u> SG - Session 11 XP - Session 11 TG- Session 11
14	EJB– TL14	All the topics as listed below from Session 12 of <i>Jakarta Enterprise Beans in Action</i> book should be covered in this session. <u>Session 12 – Jakarta Context and Dependency Injection (CDI)-II</u> ➤ Explain features of Qualifiers and Injecting Beans ➤ Describe how to use Scopes ➤ Explain how to apply Setter and Getter Methods ➤ Explain constructing a Managed Bean in a Jakarta Facelet Page ➤ Describe how to configure a CDI Application	<u>Jakarta Enterprise Beans in Action</u> SG - Session 12 XP - Session 12 TG- Session 12
15	EJB– TL15	The Try It Yourself questions of Session 11 and Session 12 of <i>Jakarta Enterprise Beans in Action</i> book should be covered in this session.	<u>Jakarta Enterprise Beans in Action</u> Session 11 & 12
16	EJB– TL16	All the topics as listed below from Session 13 of <i>Jakarta Enterprise</i>	<u>Jakarta Enterprise Beans in Action</u>

Sess No.	Session Title	Session Details	Deliverables' Mapping
		<i>Beans in Action</i> book should be covered in this session. <u>Session 13 – CDI Beans</u> ➤ Explain how to create CDI Bean Qualifiers ➤ Explain how to create Producers, Disposers, Interceptors, Events, and Stereotypes ➤ Describe Jakarta EE Declarative and Programmatic Approach ➤ Explain Security and Configure Authentication ➤ Describe basics of using Application Roles, Security Constraints, and Login Modules	SG - Session 13 XP - Session 13 TG- Session 13
17	EJB– TL17	All the topics as listed below from Session 13 and Session 14 of <i>Jakarta Enterprise Beans in Action</i> book should be covered in this session. <u>Session 14 – Packaging Jakarta Applications</u> ➤ Describe Packaging Enterprise Beans ➤ Explain Packaging Entities ➤ Elaborate Packaging Web Archives	<u>Jakarta Enterprise Beans in Action</u> SG - Session 14 XP - Session 14 TG- Session 14
18	EJB– TL18	The Try It Yourself questions of Session 13 and Session 14 of <i>Jakarta Enterprise Beans in Action</i> book should be covered in this session.	<u>Jakarta Enterprise Beans in Action</u> Session 13 & 14

6. Library References

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| ➤ Jakarta EE for Java Developers: Build Cloud-Native and Enterprise Applications Using a High-Performance Enterprise Java Platform by Rhuan Rocha |
| ➤ Java EE to Jakarta EE 10 Recipes : A Problem-Solution Approach for Enterprise Java by Josh Juneau & Tarun Telang |

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